Lab 4: Data pipeline and quality

Reading and Learning Outcomes

Reading: Reddi, Chapter 5 AI Workflow: §§5.2–5.5 (Lifecycle Stages, Problem Definition, Data Collection, Model Development); and Chapter 6 Data Engineering: §§6.2–6.8 (Problem Definition, Pipeline Basics, Data Sources, Data Ingestion, Data Processing, Data Labeling, Data Storage).

After completing this lab, students should be able to:

- Build a reproducible, leakage-free data pipeline for mixed numerical and categorical data.
- Define and execute schema, range, missing-value, duplicate, and label-quality checks.
- Engineer and evaluate derived features without contaminating the test set.
- Produce a processed-data hash, configuration, and complete data card.

1. [15] Dataset: UCI Adult via `fetch_openml(name='adult', version=2, as_frame=True)`. Write a half-page problem definition: inputs = 14 features, output = income >50K, success metric = F1 and accuracy, constraints = no leakage, reproducible pipeline.
2. [15] Ingestion step reads raw data without modifying it. Log row count 48842 and column count 15.
3. [20] Validation step. Define this schema:
   - Types: age int, workclass category, fnlwgt int, education category, education-num int, marital-status category, occupation category, relationship category, race category, sex category, capital-gain int, capital-loss int, hours-per-week int, native-country category, income category.
   - Ranges: age 17–90, hours 1–99, capital-gain 0–99999, capital-loss 0–4356.
   - Null policy: `'?'` means missing. Allowed categories include `'?'`.
   - Run checks. Report violations. Do not silently drop rows.
4. [20] Cleaning and transformation:
   - Split train/test 70/30 stratified before cleaning.
   - Fit imputer on train only. Mode for categorical, median for numeric.
   - Outliers: clip numeric features to 1st/99th percentile from train.
   - Duplicates: drop exact duplicates from train only. Keep test untouched.
   - Save fitted transformer with `joblib.dump` to `results/preprocessor.joblib`.
5. [10] Engineer at least two derived features:
   - `capital_net = capital-gain - capital-loss`
   - `hours_per_age = hours-per-week / (age - 16)`

Test with LogisticRegression baseline. Record accuracy and F1 with and without.

6. [15] Label-quality check. Draw 50 random rows. For Adult, do a duplicate-conflict audit: find duplicate feature rows with conflicting income labels. Report count. Also manually re-check 50 rows against the data dictionary for impossible values. If you use a subjective-label dataset instead, hide labels, wait 24 hours, relabel, and compute Cohen's kappa.
7. [20] Compute SHA256 hash of processed dataset. Store with config YAML. Write `data_card.md`: source, license, size, known biases, privacy concerns. Make pipeline runnable with one command.

Graded evidence: validation report, leakage-free pipeline, data card, one-command reproducibility.